

Student Name: Rob Sligter Student Number: s5564875

Section 3: Required Plots

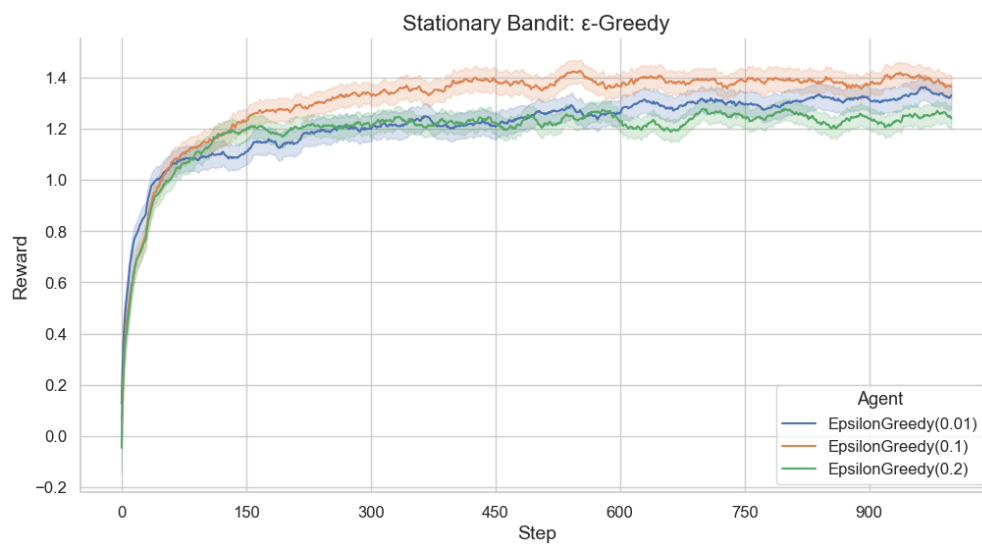
Plot 1: ϵ -Greedy Comparison

Figure 1: Comparison of ϵ -greedy with $\epsilon \in \{0.01, 0.1, 0.2\}$ on average reward.

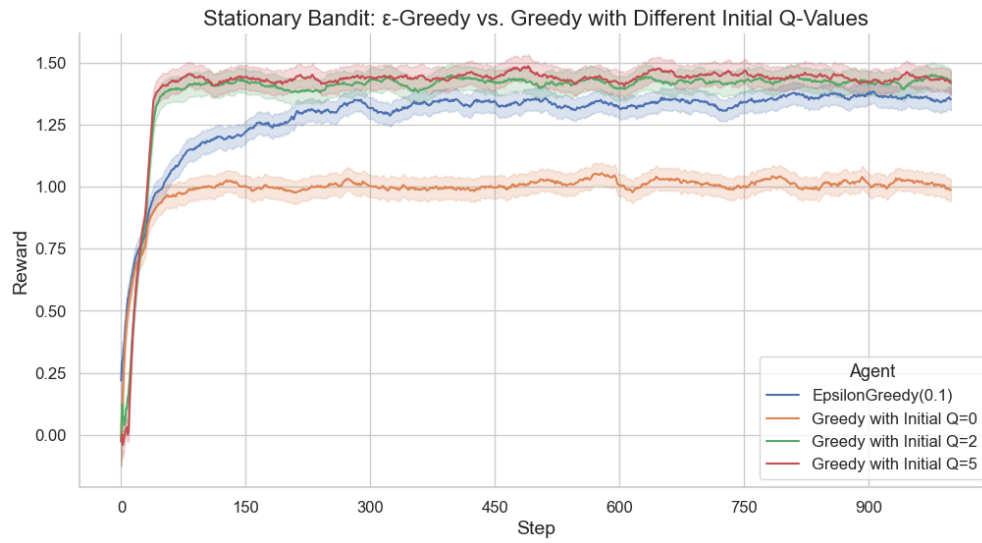
Plot 2: Optimistic Initialization Comparison

Figure 2: Comparison of ϵ -greedy ($\epsilon = 0.1$) and greedy with optimistic initialization ($Q_0 \in \{0, 2, 5\}$).

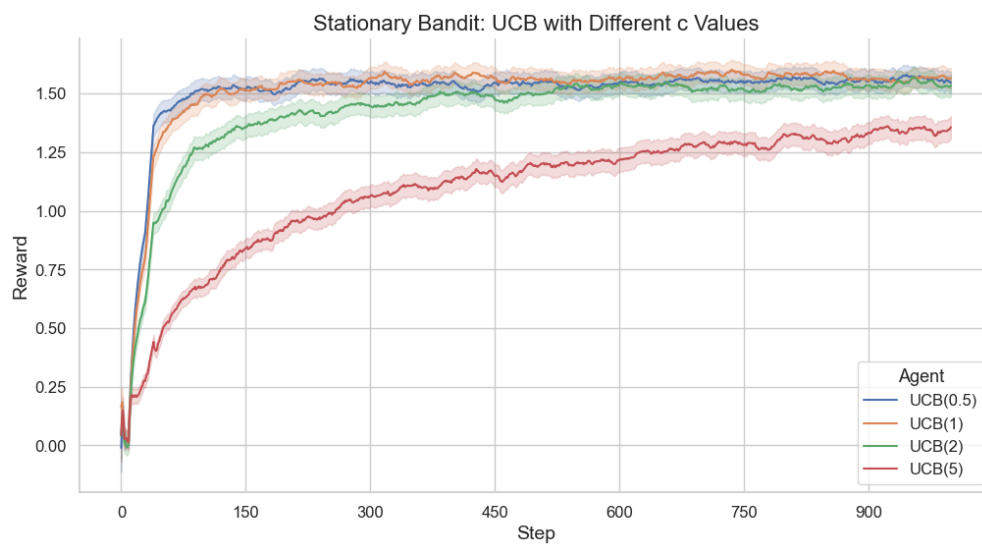
Plot 3: UCB Comparison

Figure 3: Comparison of UCB with $c \in \{0.5, 1, 2, 5\}$ on average reward.

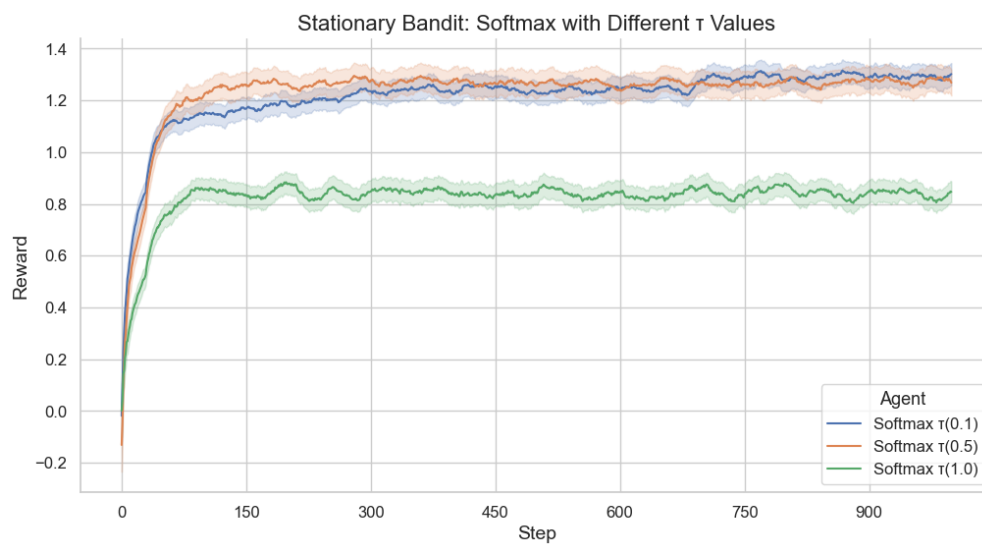
Plot 4: Softmax Comparison

Figure 4: Comparison of softmax with $\tau \in \{0.1, 0.5, 1.0\}$ on average reward.

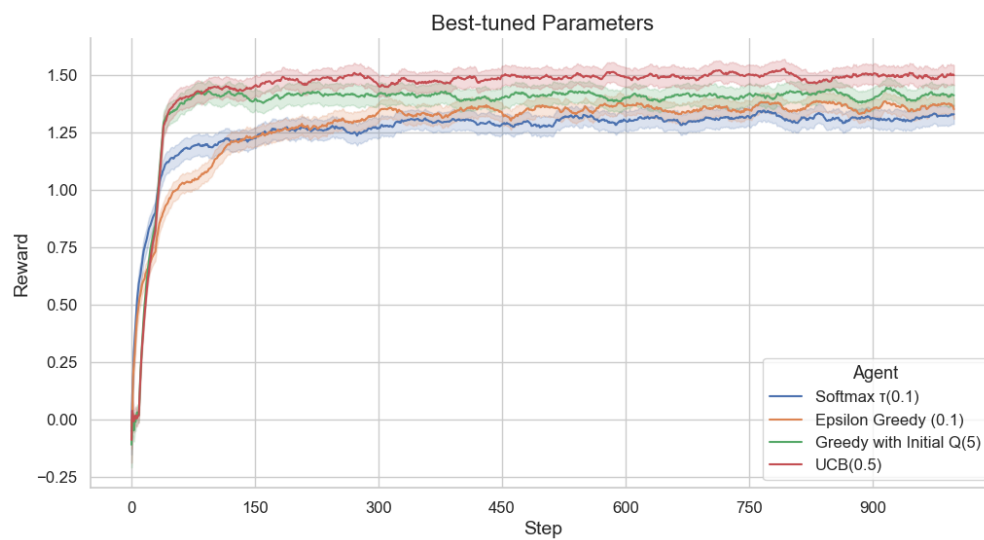
Plot 5: Overall Comparison

Figure 5: Overall comparison of ϵ -greedy, UCB, and softmax using best-tuned parameters.

Section 4: Questions

1. (ε -Greedy) How does increasing ε affect exploration and long-term reward?

Solution: Increasing ε raises the chance of random exploration, so the agent tries more arms instead of exploiting the best-known option. This prevents the agent from fully exploiting good actions, which lowers long-term rewards. As seen in Figure 1, $\varepsilon = 0.1$ balances exploration and exploitation better than higher values but still explores enough, unlike the value 0.01. Which does not explore enough slowing convergence.

2. (Optimistic Initialization) How do larger initial Q_0 values influence early exploration?

Solution: As seen in Figure 2, agents with higher initial Q_0 values like $Q_0 = 2$ or $Q_0 = 5$ converge more quickly to high rewards. This happens because Greedy always picks the best current estimate, and with all estimates starting high, the agent is forced to explore each arm until the values drop toward the true distribution. If the initial estimates start too low, the agent does not explore enough and risks missing the best arms.

3. (UCB) How do small vs. large values of c compare to ε -Greedy? Which c gives the best balance?

Solution: Smaller values of c are quicker to converge into a high reward. This is because c the uncertainty factor. A smaller c causes the model to scale the uncertainty factor down and start exploiting more. In this scenario this leads to a higher reward. As seen in figure 3 a lower c value performs well.

4. (Softmax) How does τ control the trade-off between greedy and exploratory actions?

Solution: A higher τ value makes the softmax distribution flatter, so all actions have more similar probabilities, encouraging exploration. A lower τ value sharpens the distribution, making the greedy (highest-Q) action much more likely. As seen in Figure 4, too high a τ causes excessive exploration, slowing convergence.

5. (Comparison) Which strategy performs best overall? Which one is most robust to parameter choice?

Solution: The UCB($c = 0.5$) strategy performed the best overall. Small increases in ε led to fluctuations (Figure 1), and larger c values in UCB caused too much exploration (Figure 3). The same pattern is seen in Figure 4, where higher τ values in Softmax slowed convergence. I conclude that Greedy with larger initial Q_0 values is the most robust to parameter changes, since there was little difference between $Q_0 = 2$ and $Q_0 = 5$.

6. (Reflection) If you were faced with a new, unknown bandit problem, which method would you choose? Justify your choice by referring to both your experimental results and theoretical considerations.

Solution: I would choose a Greedy strategy with larger initial Q_0 values. This starts off by exploring all arms because of their high initial Q_0 . Over time, the agent updates the estimates to get closer to the true means. In my testing, this also proved very effective. It balances exploration and exploitation naturally.